File I/O in python

python can be used to perform operations on a file. (read & write data)

Types of all files

1. Text files : txt, .docx, .log etc. :- store data characters form
2. binary files : mp4, .mov, .png, .jpeg etc.

open, read & close file

we have to open a file before reading or writing

```
f = open("file_name", "mode")
```

sample.txt
demo.docx

r: read mode
w: write mode

```
data = f.read()
f.close
```

ex:-
```
f = open("demo.txt", "r")
data = f.read()
print(data)
f.close()
```

Character	meaning
"r"	open for reading (default
'w'	open for writing, truncating the file first
'x'	create a new file and open it for writing
'a'	open for writing, appending to the end of the file if it ex
'b'	binary mode
't'	text mode (default)
'+'	open a disk file for updating (reading and writing)

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Reading a file

- data = f.read # reads entire line
  eg :- data = f.read(5)
  output :- I am

- data = f.readline() # reads one line at a time.
  print(~~data 1~~)
  eg :- line 1 = f.readline()
  print(line 1)

writing to a file

f = open("demo.txt", "w")
f.write("This is a new line") # overwrites the entire file

note :- w :- it means overwrite
a :- it means add at the end.

note :- diff :-
r+ :- read and overwrite (ptr start) - no truncate
w+ :- ~~read~~ read and overwrite truncate
a+ :- read and append (ptr end - no truncate.

with syntax :
with open("demo.txt", "a") as f :
data = f.read()
print(data)

note :- with automatically close the file, it is not necessary to write f.close in this.

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Deleting a file

using the os module.

Module (like a code library) is a file written by another programmer that generally has a functions we can use.

```
import os
os.remove(filename)
```

Q create a new file "practice.txt" using python. Add the following data in it.

```
with open("practice.txt", "w") as f:
    f.write("I am learning a python")
```

Q WAF that replaces all occurences of "java" with python" in above file.

```
with open("practice.txt", "r") as f:
    data = f.read()
    new_data = data.replace("java", "python")
    print(new_data)
with open("practice.txt", "w") as f:
    f.write(new_data)
```

Q search if the word "learning" exist in the file or not?

```
word = "learning"
with open("practice.txt", "r") as f:
    data = f.read()
    if (data.find(word) != -1):
        print("found")
    else:
        print("not found")
```

function

```
def check_for_word():
    "same as"
```

or

Check for words

Q WAP to find in which line of the file does the word "learning" occur first print -1 if word not found.

```
def check_for_line():
    word = "learning"
    data = True
    line_no = 1
    with open("practice.txt", "r") as f:
        while data:
            data = f.readline()
            if(word in data):
                print(line_no)
                return
            line_no += 1

    return -1
print(check_for_line())
```

Q. From a file containing no. seperated by comma, print the count of even numbers

```
count = 0
with open("practice.txt", "r") as f:
    data = f.read()
    print(data)
    num = data.split(",")
    print(num)
    for val in num:
        if(int(val) % 2 == 0):
            count += 1
print(count).
```

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